

PROVISIONAL PATENT APPLICATION

Constraint-Governed Computational Systems and Representation Architectures

Field of the Invention

The present disclosure relates to computational systems, artificial intelligence, biological modeling, and structured representations in which computation is governed by constraint-based transformations within structured state spaces.

Background

Conventional computational systems rely on symbolic manipulation, binary logic, and externally applied constraints. These systems generate invalid states requiring post-processing, resulting in inefficiencies and instability. Existing architectures separate representation, constraint, and computation, preventing intrinsic enforcement of valid state transitions.

Summary

The present disclosure provides systems and methods in which computation is performed through traversal of a structured state space governed by intrinsic or implicit constraints. Transformations define computation, and valid outputs arise directly from constraint-consistent transitions.

Technical Improvement

The disclosed systems improve computational efficiency by reducing invalid state exploration, improve reliability by enforcing constraint-consistent transitions, and improve generalization by embedding structure directly into computation.

System Overview

The system comprises a structured state space, a constraint structure, transformation operators, and a computation engine. Computation is performed by traversing the state space subject to constraints.

Structured State Space

In various embodiments, the structured state space may include geometric, graph-based, vector-based, manifold-based, or hybrid representations. States may be defined relationally or by coordinates.

Constraint Structure

The constraint structure may define adjacency, symmetry, topology, or equivalence relationships. Constraints may be explicit, implicit, learned, probabilistic, or emergent.

Transformation Operators

Transformation operators may include rotation, reflection, permutation, translation, scaling, or learned transformations that transition between states.

Computation Engine

A computation engine performs computation by applying transformation operators to traverse the state space subject to the constraint structure.

Example Embodiments

Tetrahedral System: States arranged in tetrahedral form with valid transitions along edges.

Artificial Intelligence System: Representations correspond to positions in a structured state space with constraint-governed transitions improving generalization.

Biological System: Codons mapped into structured state space, mutations correspond to transitions.

Processor System: Multi-state computational elements performing transformation-based computation.

Hybrid System: Symbolic inputs mapped to geometric representations and processed via constraint-based transformations.

Example Algorithm

One example implementation is provided below, though many variations are possible:

```
state = initial_state
for op in operations:
    if op in valid_transitions(state):
        state = apply(op, state)
return state
```

Distributed Systems

The system may be distributed across multiple computing components that collectively perform computation.

Variations and Extensions

In some embodiments, variation 1 may include differences in state space, constraints, or transformations.

In some embodiments, variation 2 may include differences in state space, constraints, or transformations.

In some embodiments, variation 3 may include differences in state space, constraints, or transformations.

In some embodiments, variation 4 may include differences in state space, constraints, or transformations.

In some embodiments, variation 5 may include differences in state space, constraints, or transformations.

In some embodiments, variation 6 may include differences in state space, constraints, or transformations.

In some embodiments, variation 7 may include differences in state space, constraints, or transformations.

In some embodiments, variation 8 may include differences in state space, constraints, or transformations.

In some embodiments, variation 9 may include differences in state space, constraints, or transformations.

In some embodiments, variation 10 may include differences in state space, constraints, or transformations.

In some embodiments, variation 11 may include differences in state space, constraints, or transformations.

In some embodiments, variation 12 may include differences in state space, constraints, or transformations.

In some embodiments, variation 13 may include differences in state space, constraints, or transformations.

In some embodiments, variation 14 may include differences in state space, constraints, or transformations.

In some embodiments, variation 15 may include differences in state space, constraints, or transformations.

Generalization and Scope

The systems and methods described herein are not limited to any specific implementation and may be applied across computational, artificial intelligence, biological, physical, or hybrid systems.

Any system in which relationships between states define permissible transitions and computation is performed through traversal of such relationships falls within the scope of this disclosure.

Expanded Variations

In some embodiments, the constraint structure is represented as a probabilistic distribution over permissible transitions.

In some embodiments, constraints are generated or modified by a neural network or machine learning model.

In some embodiments, the structured state space is continuous rather than discrete.

In some embodiments, the structured state space is hierarchical, comprising multiple levels of abstraction.

In some embodiments, transformation operators are learned dynamically during operation.

In some embodiments, constraint enforcement is approximate rather than exact.

In some embodiments, constraints emerge from system dynamics rather than explicit specification.

In some embodiments, state transitions are governed by energy-based models.

In some embodiments, multiple structured state spaces interact to form a composite system.

In some embodiments, transitions may be reversible or irreversible.

In some embodiments, the system includes stochastic elements influencing transitions.

In some embodiments, constraint structures vary over time.

In some embodiments, state spaces are embedded within higher-dimensional representations.

In some embodiments, constraints are encoded as adjacency matrices or tensors.

In some embodiments, computation is distributed across multiple interacting state spaces.

In some embodiments, transitions are governed by learned policies.

In some embodiments, the system integrates symbolic and subsymbolic representations.

In some embodiments, constraints are partially observed or inferred.

In some embodiments, transformation operators include nonlinear mappings.

In some embodiments, the system operates in real-time environments.

In some embodiments, constraints enforce safety or stability conditions.

In some embodiments, the structured state space represents physical systems.

In some embodiments, computation is performed in parallel across multiple states.

In some embodiments, constraints encode domain-specific rules.

In some embodiments, state transitions correspond to semantic transformations.

In some embodiments, the system adapts constraints based on feedback.

In some embodiments, multiple constraint structures are combined.

In some embodiments, transformation operators are composable.

In some embodiments, the system includes reinforcement learning components.

In some embodiments, state spaces evolve over time.